

Real-Time Peak-Demand Minimization with Energy Storage Using Competitive Ratio [★]

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Abstract

This paper studies minimizing the peak demand for large-load users with energy storage by making irrevocable storage-consuming decisions with unknown future demands in a daily on-peak horizon. We use competitive analysis to optimize online decision-making for robustness and fairness over all possible input sequences. The best competitive ratio (CR) for the online problem is computed by solving a linear number of linear-fractional programs and used to obtain an optimal online strategy, which is a static feedback policy affine over a synthetic state. Further, we generalize the concept of CR to shrinking-horizon CRs in response to real-time inputs and decisions. Accordingly, a *real-time optimal* online algorithm is developed to achieve the rolling-optimized shrinking-horizon CRs. This algorithm is a dynamic feedback policy, retains the global optimal worst-case performance, and improves the general-case performance. Trace-driven simulations show that the real-time optimal algorithm can significantly decrease the peak demand compared to baseline alternatives under typical settings.

Key words: Optimization under uncertainties, Resource allocation, Receding horizon control, Demand response

1 Introduction

Apart from energy pricing schemes like time-of-use (ToU) [2] and consumption-based block (CBB) [3], utilities introduce the demand charge, which penalizes the maximum power (demand) to incentivize consumers to reduce their peak power consumption [4–6]. The demand charge often accounts for a significant portion of a bill, up to 90% [7]. Many studies have focused on flattening the aggregated demand for a collection of loads like heat

pumps and electric vehicles (EVs) [8–12]. However, the efforts of a single user to shave peak demands should not be overlooked [4, 5, 13, 14]. Therefore, our work explores a scenario where a large-load user, such as a data center and an EV charging station, aims to minimize the peak demand using in-house energy storage.

A common practice in industry is that a user replenishes energy storage in daily off-peak hours when electricity prices are low, and then strategically discharges the storage during on-peak hours when the electricity is dear [2]. Thus, the optimal online decision-making problem is to use the fixed storage capacity to minimize the peak demand each day. This problem is challenging for the following reasons. First, the user must make irrevocable storage-consuming real-time decisions with unknown future information [15, 16]. Second, the storage capacity renders storage-consuming decisions over time interdependent. Finally, a peak load committed previously is not reducible, complicating real-time decision-making. This problem also finds applications in cloud computing and supply chain management [17].

Our work contributes to validating competitive analysis as an effective tool for addressing the online peak-demand minimization (PDM) problem using energy

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storage consumption. Competitive analysis [18] is a popular tool for robustness in power [19–23] and control [24–29] applications. *Competitive ratio* (CR) is used as the performance metric. Similar to (dynamic) regret [8, 30], CR stresses the relative performance of an online/real-time algorithm with little future information compared to an offline optimization in hindsight [18, 24], where full information on net loads is available. Thus, CR is concerned with the robustness and fairness of an online algorithm via the worst-case online-to-offline performance ratio over all possible input sequences.

The primal-dual [31] and primal-based [22, 32] approaches in the competitive analysis are both useful for online allocation of an inventory without replenishment to optimize a sum objective function $\sum_t f_t(x_t)$, like the classic one-way trading problem [33]. The primal-based approach is particularly advantageous when considering a minimax objective $\min \max_t f_t(x_t)$ [17, 19, 34]. For the online PDM problem, a primal-based algorithmic framework called CR-Pursuit [17, 19, 22, 32, 34] is applied to achieve optimal robustness regarding CR. This framework is further generalized by developing a real-time optimal scheme for improved efficiency. The new approach is inspired by the rolling optimization framework in model predictive control (MPC) [5, 35] and pursues robustness and efficiency by optimizing worst-case relative performances using real-time information [36, 37].

The following is the organization of the rest of the paper, along with the main contributions.

- The best CR for the online PDM problem, formulated in Sec. 2, is computed by solving a linear number of linear-fractional programs in Sec. 3. Moreover, an optimal online algorithm is proposed by using the best CR and solving evolving offline optimization problems. To the best of our knowledge, these results are among the first (and optimal) for this practically relevant and challenging online decision-making problem regarding CR.

- In Sec. 4, we generalize the concept of CR to shrinking-horizon CRs in response to real-time information. Accordingly, a *real-time optimal* online algorithm is developed to adapt to the rolling-optimized competitiveness regarding the residual uncertainty and real-time information. The algorithm retains the best CR and attains improved general-case performance, relative to the optimal online algorithm obtained in Sec. 3.

- The algorithms are evaluated via real-world traces from an EV charging station in Sec. 5. Under typical settings, the real-time optimal algorithm outperforms baseline alternatives, including the shrinking receding horizon control (RHC) [35], with a quarter of the operation horizon of accurate predictions.

Finally, the paper is concluded with challenges and future directions in Sec. 6. The nomenclature is given in

Table 1 for easy reference. Note that a minimization problem and its maximization counterpart require distinct competitive analyses [18]. Thus, this work focusing on PDM differs from and advances our previous related work for peak-reduction maximization (PRM) [34].

Table 1
Nomenclature

T	Number of time slots of the on-peak period
$\llbracket T \rrbracket$	Set $\{1, 2, \dots, T\}$ for $T \in \mathbb{N}$
$[x]^+$	Maximum of x and 0 for $x \in \mathbb{R}$
$\lfloor x \rfloor$	Largest integer no more than $x \in \mathbb{R}$
$\mathcal{X} \setminus \mathcal{Y}$	$\{x \mid x \in \mathcal{X} \text{ \& } x \notin \mathcal{Y}\}$ for two sets $\mathcal{X}$ and $\mathcal{Y}$
c	Scaled capacity of energy storage, (kWh)
$\bar{\delta}$	Maximum consuming amount per slot, (kWh)
$\underline{d}, \bar{d}$	Net load lower and upper bounds, (kWh)
$\mathbf{d}$	Net load profile $[d_1 \ \dots \ d_t \ \dots \ d_T]' \in \mathbb{R}^T$
d_t^m	Realized value of d_t , (kWh)
δ_t	Decision variable at time slot t , (kWh)
δ_t^m	Realized value of δ_t , (kWh)
v^{OPT}	Optimal offline peak demand function
v^{ON}	Online peak demand function
$\Delta_t^\bullet$	Storage-consuming function at time slot t
$\mathbf{d}_t$	$[d_1^m \ d_2^m \ \dots \ d_t^m \ \underline{d} \ \dots \ \bar{d}]' \in \mathbb{R}^T$
π^*	Best CR for online PDM
π_t^*	Optimal shrinking-horizon CR at time slot t

2 System model and problem formulation

2.1 System model

Evenly divide a daily on-peak horizon into T time slots by the power pricing interval (e.g., 30 minutes). Accordingly, the power demand (in kW) is converted into energy consumption (in kWh) of a slot.

Constraints due to energy storage consumption.

Let c and $\bar{\delta}$ (in kWh) denote the energy storage capacity scaled by the discharging efficiency and the maximum possible amount of storage that can be allocated in a time slot, respectively. To avoid triviality, assume that the maximum discharge rate will exhaust the capacity, i.e., $T\bar{\delta} > c$. Let δ_t be the amount of energy storage to be consumed at time slot t . The storage-consuming solution should satisfy the capacity constraint

$$\sum_{t=1}^T \delta_t \leq c, \quad (1)$$

and the rate constraint

$$0 \leq \delta_t \leq \bar{\delta}, \ \forall t \in \llbracket T \rrbracket. \quad (2)$$

Uncertain system input sequence due to net load.

The net load profile $\mathbf{d} = [d_1, \dots, d_T]' \in \mathbb{R}^T$ represents the difference between the gross demand and the local renewable generation, where d_t gives the net load at time slot $t \in \llbracket T \rrbracket$. The gross demand and renewable generation are hard to predict, so we only assume to know beforehand that the net load at each time slot is within the uniform lower and upper bounds $\underline{d} \in \mathbb{R}^+$ and $\bar{d} \in \mathbb{R}^+$. Thus, the uncertainty set of the system input sequence is

$$\mathcal{D} \triangleq \{\mathbf{x} \in \mathbb{R}^T \mid \underline{d} \leq x_t \leq \bar{d}, \forall t \in \llbracket T \rrbracket\}.$$

Note that it is trivial to define the uncertainty set with time-dependent bounds.

Peak demand. Let d_t^m be the measured value of d_t and δ_t^m be the realized storage-consuming amount at time slot t . Then, the peak demand is given by

$$\max_{t \in \llbracket T \rrbracket} (d_t^m - \delta_t^m).$$

2.2 Clairvoyant offline peak-demand minimization

Before proceeding, we detour to study the clairvoyant setting where the storage-consuming solution can be given in hindsight with the perfect information of the net load profile $\mathbf{d}$. This unrealistic setting is needed to define CR and serves as the baseline for developing CR-based online decision-making methods. The offline storage-discharging PDM problem is

$$\begin{aligned} \text{Offline PDM: } & \min_{\delta \in \mathbb{R}^T} \max_{t \in \llbracket T \rrbracket} (d_t^m - \delta_t) \\ \text{subject to } & (1); (2); \\ & 0 \leq \delta_t \leq d_t^m, \forall t \in \llbracket T \rrbracket, \end{aligned} \quad (3)$$

which minimizes the peak demand over the whole horizon $\llbracket T \rrbracket$ after $\mathbf{d}$ is completely measured. In this case, minimizing the peak demand is equivalent to maximizing the peak-demand reduction.

The following proposition states the optimal objective value v^* and solution δ^* to the offline PDM problem.

Proposition 1 *Given the realization of $\mathbf{d}$, there exists u with $\sum_{t=1}^T [d_t^m - u]^+ = c$. Moreover, the optimal offline peak demand and storage consumption are, respectively,*

$$v^* = \max \left\{ 0, u, \max_{t \in \llbracket T \rrbracket} d_t^m - \bar{\delta} \right\} \text{ and } \delta_t^* = [d_t^m - v^*]^+.$$

Proposition 1 is proven in Appendix A. It admits a bisection solution for v^* after $\{d_t^m\}_{t=1}^T$ are known. Thus, we define the optimal offline peak demand function

$$\mathbf{x} \in \mathcal{D} \mapsto v^{OPT}(\mathbf{x}) \in \mathbb{R},$$

where $v^{OPT}(\mathbf{x})$ is the optimal offline peak demand when the realization of the net load profile $\mathbf{d}$ equals $\mathbf{x}$. This optimal offline peak demand function will facilitate the implementation of the proposed online algorithms.

2.3 Online decision-making and competitive ratio

Since d_t is not available until the current time t in real time, an online algorithm for PDM has to irrevocably determine the consuming amount δ_t at time slot t based on the available net loads up to t . The online peak minimization is subject to the remaining capacity and the rate limit $\bar{\delta}$

$$c_t = c - \sum_{k=1}^{t-1} \delta_k^m.$$

As a new net load d_t is measured d_t^m at time slot t , the current decision δ_t has to satisfy the online constraint

$$0 \leq \delta_t \leq \min\{\bar{\delta}, d_t^m, c_t\}, \quad (4)$$

while making the peak demand as close as possible to the optimal offline peak demand that can only be obtained in hindsight. Thus, an online PDM algorithm $\mathfrak{A}$ is a sequence of functions $\{\Delta_t^{\mathfrak{A}}\}_{t=1}^T$, where

$$\mathbf{x} \in [\underline{d}, \bar{d}] \mapsto \Delta_t^{\mathfrak{A}}(\mathbf{x}; \{d_k^m\}_{k=1}^{t-1}, \{\delta_k^m\}_{k=1}^{t-1}) \in [0, \bar{\delta}],$$

such that the storage-consuming decision δ_t at time slot t is determined by the function $\Delta_t^{\mathfrak{A}}$ and respects the current online constraint (4), that is,

$$\delta_t = \Delta_t^{\mathfrak{A}}(d_t^m; \{d_k^m\}_{k=1}^{t-1}, \{\delta_k^m\}_{k=1}^{t-1}) \leq \min\{\bar{\delta}, d_t^m, c_t\},$$

for all possible net load profile realizations in $\mathcal{D}$.

Moreover, the online peak demand function is defined as

$$(\mathbf{x} \in \mathcal{D}, \{\Delta_t^{\mathfrak{A}}\}_{t=1}^T) \mapsto v^{ON}(\mathbf{x}, \{\Delta_t^{\mathfrak{A}}\}_{t=1}^T) \in \mathbb{R},$$

where $v^{ON}(\mathbf{x}, \{\Delta_t^{\mathfrak{A}}\}_{t=1}^T)$ is the peak demand under the algorithm $\mathfrak{A}$ when the realization of $\mathbf{d}$ equals $\mathbf{x}$.

Finding an online PDM algorithm is easy because the online constraint (4) can be readily verified. The challenge lies in designing an online PDM algorithm with provable performance against the uncertainty $\mathcal{D}$ in net loads and bridging the gap between online and clairvoyant offline decision-making. To this end, we consider the online PDM problem in the sequel via competitive analysis for robustness and efficiency.

First of all, the performance metric for an online PDM algorithm is defined. Competitive ratio is a worst-case multiplicative analog of dynamic regret [27, 38]. The CR of an online PDM algorithm $\mathfrak{A}$ is defined as the worst-case ratio between the peak demand under the algorithm $\mathfrak{A}$ and the optimal offline peak demand for the

same realization of the net load profile, over all possible realizations in $\mathcal{D}$

$$\text{CR}_{\mathfrak{A}} = \max_{\mathbf{x} \in \mathcal{D}} \frac{v^{ON}(\mathbf{x}, \{\Delta_t^{\mathfrak{A}}\}_{t=1}^T)}{v^{OPT}(\mathbf{x})}.$$

By definition, the CR in competitive analysis indicates a two-player game between an online algorithm and an adversary that can specify the input sequence or the realization of the net load profile. The CR is always no less than one, and a smaller CR means better competitiveness against $\mathcal{D}$. The optimal online PDM algorithm has the smallest CR among all online PDM algorithms, called the best CR π^* . The best CR π^* is a property of the online PDM problem, quantifying the fundamental gap between online decision-making and the clairvoyant offline optimum due to the uncertainty $\mathcal{D}$ [18]. Thus, the best CR π^* can be pre-computed after knowing $\mathcal{D}$ but observing none of the net loads.

To avoid $\pi^* = \infty$, the following assumption is needed.

Assumption 1 $v^{OPT}(\mathbf{x}) > 0$ for all $\mathbf{x} \in \mathcal{D}$.

Remark 1 The raw values of $\underline{d}$, $\bar{d}$, and d_t^m for all $t \in \llbracket T \rrbracket$ can be lifted by a constant until Assumption 1 is satisfied.

3 Online PDM with specified competitive ratios

3.1 CR-Pursuit framework

Finding an online algorithm of the best CR π^* is generally challenging because it requires optimization in the functional space. However, the CR of an online algorithm is a scalar. If we can associate each ratio that is no less than the best CR π^* with an online algorithm whose CR is the ratio, then it suffices to search a one-dimensional space for the optimal online algorithm. This idea has led to the development of a primal-based online algorithmic framework in competitive analysis, called CR-Pursuit [17, 19, 22, 32, 34].

This framework makes online decisions with the goal of “pursuing” a given scalar to be the CR of the resulting online algorithm. The method for pursuing the given scalar is problem-specific. Usually, a set of scalars are pursuable via the same method, leading to a collection of online algorithms. If the smallest pursuable scalar is also the best CR for the online optimization problem, the associated CR-Pursuit method helps significantly reduce the search space for the optimal online algorithm to a one-dimensional space over the scalars. Next, we propose a CR-Pursuit method for the online PDM.

3.2 Online PDM using the best CR π^*

At time slot t , the shrunk uncertainty set is

$$\mathcal{D}_t = \{\mathbf{x} \in \mathcal{D} \mid x_k = d_k^m, \forall k \in \llbracket t \rrbracket\},$$

and a reference input sequence $\mathbf{d}_t$ can be constructed based on the current information:

$$\mathbf{d}_t = [d_1^m \ d_2^m \ \cdots \ d_t^m \ \underline{d} \ \cdots \ \bar{d}]' \in \mathcal{D}_t.$$

Then, the value $v^{OPT}(\mathbf{d}_t)$ represents the optimal offline peak demand, assuming that the realized net load profile is $\mathbf{d}_t$. We highlight that $v^{OPT}(\mathbf{d}_T)$ corresponds to the optimal peak demand for the true offline PDM problem. For each online algorithm whose CR is no more than π , the online-to-offline objective ratio under the algorithm with $\mathbf{d}$ realized by $\mathbf{d}_t$ does not exceed π . Thus, the storage-consuming amount δ_t should at least be

$$[d_t^m - \pi v^{OPT}(\mathbf{d}_t)]^+.$$

Following the above analysis, we define a class of algorithms, each parameterized by a given scalar $\pi \geq 1$ and named pCR-PDM(π). The pseudocodes are presented in Algorithm 1. Each pCR-PDM(π) is characterized by a sequence of functions $\{\Delta_t^\pi\}_{t=1}^T$, where

$$\Delta_t^\pi(d_t^m; \{d_k^m\}_{k=1}^{t-1}, \{\delta_k^m\}_{k=1}^{t-1}) = [d_t^m - \pi v^{OPT}(\mathbf{d}_t)]^+.$$

The question lies in whether the pCR-PDM(π) algorithm respects the capacity and rate constraints (1)-(2) for all possible realizations of $\mathbf{d}$. If the answer is YES, then pCR-PDM(π) is an online PDM algorithm, and its CR is upper bounded by π . The proposition below characterizes the feasibility of pCR-PDM(π) by a threshold condition.

Proposition 2 Given $\pi \in \mathbb{R}$, pCR-PDM(π) is feasible for the online PDM if and only if

$$\Phi(\pi) \triangleq \max_{\mathbf{x} \in \mathcal{D}} \sum_{t=1}^T \Delta_t^\pi(x_t; \{d_k^m\}_{k=1}^{t-1}, \{\delta_k^m\}_{k=1}^{t-1}) \leq c.$$

Proposition 2, proven in Appendix B, suggests that the capacity constraint (1) plays a dominant role over the rate constraint (2). The function $\Phi(\pi)$ specifies the minimum amount of energy storage required to make pCR-PDM(π) an online PDM algorithm. Assumption 1 indicates that $v^{OPT}(\mathbf{d}_t) > 0$ always holds. The following lemma, proven in Appendix C, further points out the monotonicity of $\Phi(\pi)$ in π .

Lemma 1 If $\bar{\pi}$ is the smallest ratio with $\Phi(\bar{\pi}) = 0$, then $1 \leq \hat{\pi} < \bar{\pi} \leq \bar{\pi}$ implies $\Phi(\hat{\pi}) > \Phi(\bar{\pi})$.

Algorithm 1: pCR-PDM(π)

Parameters: $c, T, \bar{\delta}, [\underline{d}, \bar{d}]$;**Input:** $\pi > 1$ and a realization of $\mathbf{d}$;**Output:** A realization of δ ;

```
1 for  $t = 1, 2, \dots, T$  do
2   Construct  $\mathbf{d}_t$  from measured inputs  $\{d_k^m\}_{k \in [t]}$ ;
3    $\delta_t = [d_t^m - \pi v^{OPT}(\mathbf{d}_t)]^+$ ;
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Lemma 1 suggests that a smaller π leads to a more optimistic pCR-PDM algorithm, which, in the worst case, requires more energy storage to shave peaks as desired. If $\Phi(\pi) > c$, then pCR-PDM(π) cannot be feasible for the online PDM. It follows from Lemma 1 that the scalars associated with feasible pCR-PDM algorithms form an interval. More interestingly, the following theorem shows that the best pCR-PDM algorithm is also optimal among all online algorithms for PDM regarding CR.

Theorem 1 *Given $(c, T, \underline{d}, \bar{d}, \bar{\delta})$, the equation $\Phi(\pi) = c$ admits a unique solution, which is exactly the best CR π^* .*

Theorem 1, proven in Appendix D, allows us to reduce the search space of an optimal online PDM algorithm to a one-dimensional interval. If the best CR π^* is known, we obtain an optimal online policy – pCR-PDM(π^*). pCR-PDM(π^*) can be regarded as a static feedback policy affine over the synthetic state $v^{OPT}(\mathbf{d}_t)$. Similar to MPC [5, 35], pCR-PDM(π^*) requires solving an evolving offline optimization problem at each time slot. However, instead of applying the obtained solution for the control input, pCR-PDM(π^*) uses the obtained objective value for synthesizing a state.

Remark 2 *At time slot t , the remaining capacity c_t is one of the current states of the online PDM system and c_t appears in the online constraint (4). Nevertheless, pCR-PDM(π^*) does not rely on c_t , for all $t \in [T]$.*

Remark 3 *The value $v^{OPT}(\mathbf{d}_t)$ specifies the lowest possible optimal offline peak demand based on the current information:*

$$v^{OPT}(\mathbf{d}_t) = \min_{\mathbf{x} \in \mathcal{D}_t} v^{OPT}(\mathbf{x}).$$

Note that we use the lower limit $\underline{d}$ rather than the upper limit $\bar{d}$ to construct $\mathbf{d}_t$, ensuring that the parameter π in the pCR-PDM(π) algorithm is directly tied to the competitive ratio of the algorithm.

3.3 Computing the best CR π^*

It remains to compute the best CR π^* beforehand for implementing pCR-PDM(π^*). Deriving the analytic expression of the best CR π^* seems hard as π^* is the solution of the complicated univariate equation $\Phi(\pi) = c$ in Theorem 1. Neither is it efficient to search for π^* by dynamic

programming with a continuous decision space. As a key contribution, the computation of π^* is transformed into solving a sequence of linear-fractional problems, which is as simple as solving linear programs [39, Section 4.3.2].

First, if pCR-PDM(π) is feasible, then for each subset $\mathcal{I}$ of $[T]$, it holds that

$$\sum_{i \in \mathcal{I}} (d_i^m - \pi v^{OPT}(\mathbf{d}_i)) \leq \sum_{t=1}^T [d_t^m - \pi v^{OPT}(\mathbf{d}_t)]^+ \leq c,$$

implying

$$\pi \geq \frac{-c + \sum_{i \in \mathcal{I}} d_i^m}{\sum_{i \in \mathcal{I}} v^{OPT}(\mathbf{d}_i)};$$

Thus, the best CR π^* is lower bounded by the right-hand side (RHS) of the above inequality. The following proposition states that the lower bound can be tight for certain $\mathcal{I}$ and realization of $\mathbf{d}$.

For $t \in [T]$ and a variable $\mathbf{x} = [x_1 \ \dots \ x_T]' \in \mathbf{R}^T$, define

$$\mathbf{x}_t = [x_1 \ \dots \ x_t \ \underline{d} \ \dots \ \underline{d}]' \in \mathbf{R}^T.$$

Proposition 3 *The best CR π^* for online PDM equals*

$$\max_{\mathcal{I} \subseteq [T], \mathbf{x} \in \mathcal{D}} \frac{\sum_{i \in \mathcal{I}} x_i - c}{\sum_{i \in \mathcal{I}} v^{OPT}(\mathbf{x}_i)}.$$

Proposition 3, following from Theorem 1, suggests that the best CR π^* can be pre-computed by solving a sequence of linear-fractional problems. Each problem, referred to as CR-Compute($\mathcal{I}$), is parameterized by a nonempty subset $\mathcal{I}$ of the set $[T]$,

$$\text{CR-Compute}(\mathcal{I}) : \max_{\mathbf{x} \in \mathcal{D}, u_i, \delta_{ij}} \frac{\sum_{i \in \mathcal{I}} x_i - c}{\sum_{i \in \mathcal{I}} u_i} \quad (5)$$

$$\text{subject to } \sum_{j=1}^T \delta_{ij} = c \quad \forall i \in [T]; \quad (6)$$

$$0 \leq \delta_{ij} \leq \bar{\delta} \quad \forall i, j \in [T]; \quad (7)$$

$$x_j - \delta_{ij} \leq u_i \quad \forall 1 \leq j \leq i \leq T; \quad (8)$$

$$\underline{d} - \delta_{ij} \leq u_i \quad \forall 1 \leq i < j \leq T. \quad (9)$$

The above variables and constraints are explained as follows. For each $i \in [T]$, the variables $\{\delta_{ij}\}_{j \in [T]}$ and u_i are respectively associated with the solution and objective of the clairvoyant PDM with $\mathbf{d} = \mathbf{x}_i$. Constraints (6)-(7) are derived from the the offline PDM constraints (1)-(2). Constraints (8)-(9) ensure that u_i is no less than the optimal offline peak demand with $\mathbf{d} = \mathbf{x}_i$.

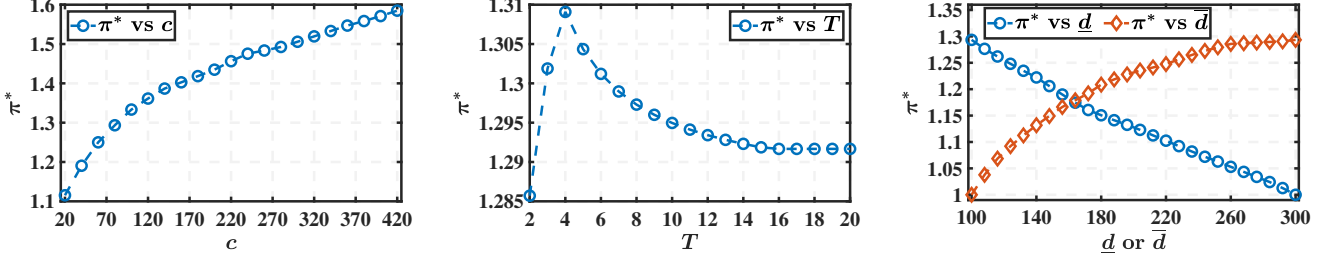


Figure 1. The best CR π^* varies as $c, T, \underline{d}$, and $\bar{d}$ with the default setting $c = 80, T = 12, \underline{d} = 100, \bar{d} = 300$ and $\bar{\delta} = \infty$.

The objective (5) comes from Proposition 3. The optimal objective value of $\text{CR-Compute}(\mathcal{I})$ depends solely on the prior information $(c, T, \underline{d}, \bar{d}, \bar{\delta})$ and the set $\mathcal{I}$. By the formulation of $\text{CR-Compute}(\mathcal{I})$, the lemma below follows.

Lemma 2 *For a nonempty set $\mathcal{I} \subseteq \llbracket T \rrbracket$, the optimal objective value of $\text{CR-Compute}(\mathcal{I})$ is*

$$\max_{\mathbf{x} \in \mathcal{D}} \frac{\sum_{i \in \mathcal{I}} x_i - c}{\sum_{i \in \mathcal{I}} v^{OPT}(\mathbf{x}_i)}.$$

With Proposition 3 and Lemma 2, the best CR π^* can be obtained by solving an exponential number (2^T) of linear-fractional programs $\text{CR-Compute}(\mathcal{I})$. Moreover, each $\text{CR-Compute}(\mathcal{I})$ can be transformed into a linear program [39, Section 4.3.2]. While these linear programs can be solved in parallel, it is desired to exclude as many redundant $\text{CR-Compute}(\mathcal{I})$ programs as possible for computational efficiency. To this end, the problem structure should be exploited.

Definition 1 *A realization of the net load profile is said to be worst-case regarding an online algorithm if applying the algorithm with the realization leads to $\sum_{t=1}^T \delta_t^m = c$.*

The following proposition points out a particular worst-case realization for $\text{pCR-PDM}(\pi^*)$.

Lemma 3 *There exists a worst-case realization regarding $\text{pCR-PDM}(\pi^*)$ such that, for certain $\tau \in \llbracket T \rrbracket$,*

$$\delta_t = \Delta_t^{\pi^*} (d_t^m; \{d_k^m\}_{k=1}^{t-1}, \{\delta_k^m\}_{k=1}^{t-1}) \begin{cases} > 0, \text{ if } t \in \llbracket \tau \rrbracket, \\ = 0, \text{ otherwise.} \end{cases}$$

Lemma 3, proven in Appendix E, shows that, for a specific worst-case input sequence, $\text{pCR-PDM}(\pi^*)$ will keep consuming stored energy until the storage runs out. Following Lemma 2, Lemma 3, and Proposition 3, the theorem below is obtained.

Theorem 2 *Given $\tau = \lfloor c/\bar{d} \rfloor$, the best CR is the maximum of the optimal objective values of the set of programs*

$\{\text{CR-Compute}(\llbracket t \rrbracket) \mid t = \tau + 1, \dots, T\}$, namely,

$$\pi^* = \max_{\mathcal{I} \in \{\llbracket t \rrbracket \mid t = \tau + 1, \dots, T\}} \max_{\mathbf{x} \in \mathcal{D}} \frac{\sum_{i \in \mathcal{I}} x_i - c}{\sum_{i \in \mathcal{I}} v^{OPT}(\mathbf{x}_i)}.$$

With Theorem 2, the number of $\text{CR-Compute}(\mathcal{I})$ programs for computing π^* is significantly reduced from an exponential (2^T) to a linear number (less than T) regarding the time-slot number T .

The best CR π^* solely depends on the prior information $(c, T, \underline{d}, \bar{d}, \bar{\delta})$. Figure 1 illustrates π^* over $(c, T, \underline{d}, \bar{d})$ with a large enough $\bar{\delta}$. Figure 1 shows that the best CR π^* decreases as the capacity decreases or the uncertainty set $\mathcal{D}$ shrinks, but the change of π^* is not monotonic as T varies. Intuitively, a larger T leads to a higher dimension of the input sequence space, making online decision-making harder. This may account for the initial increase of π^* in T . On the other hand, as T grows, the average resource per slot decreases due to the limited capacity. Thus, the relative performance between the online and the optimal offline outcomes may improve. This may explain why π^* decreases in T after T exceeds a certain value. Nevertheless, the best CR π^* will not reach one due to net load uncertainty.

4 Real-Time Optimal pCR-PDM

CR-based algorithms are often criticized for being pessimistic because they focus on the worst cases regarding the original uncertainty and do not respond to the uncertainty evolution in real time. For example, $\text{pCR-PDM}(\pi^*)$ uniformly pursues the pre-computed best CR π^* for all time slots, and the best CR is defined on the original uncertainty set $\mathcal{D}$. Note that the current uncertainty $\mathcal{D}_t$ shrinks as new observations arrive. To pursue real-time optimal robustness, the concept of (best) CR is next generalized to (optimal) shrinking-horizon CRs.

4.1 Pursuing optimal shrinking-horizon CRs

At time slot t , the net loads up to time slot t are realized $\{d_k^m\}_{k=1}^t$ and the actions up to time slot $t - 1$

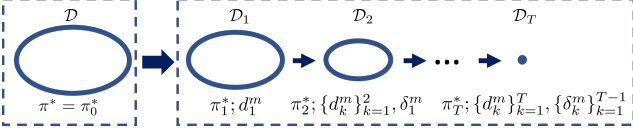


Figure 2. An illustration of extending the pCR-PDM(π^*) algorithm to the real-time optimal pCR-PDM algorithm.

are committed $\{\delta_k^m\}_{k=1}^{t-1}$. Similar to before, a shrinking-horizon online PDM algorithm $\mathfrak{A}_t$ at time slot t is a sequence of functions $\{\Delta_j^{\mathfrak{A}_t}\}_{j=t}^T$, where

$$x \in [\underline{d}, \bar{d}] \mapsto \Delta_j^{\mathfrak{A}_t} \left(x; \{d_k^m\}_{k=1}^{j-1}, \{\delta_k^m\}_{k=1}^{j-1} \right) \in [0, \bar{\delta}],$$

such that the storage-consuming decision δ_j at time slot $j \in [T] \setminus [t-1]$ is determined by the function $\Delta_j^{\mathfrak{A}_t}$ and respects the current online constraint (4), that is,

$$\delta_j = \Delta_j^{\mathfrak{A}_t} \left(d_j^m; \{d_k^m\}_{k=1}^{j-1}, \{\delta_k^m\}_{k=1}^{j-1} \right) \leq \min\{\bar{\delta}, d_j^m, c_j\},$$

for all possible net load profile realizations in $\mathcal{D}_t$.

Also, the online peak demand function is modified as

$$(x, \{\Delta_k^{\mathfrak{A}_t}\}_{k=t}^T) \mapsto v^{ON} \left(x, \{\Delta_k^{\mathfrak{A}_t}\}_{k=t}^T; \{d_k^m\}_{k=1}^t, \{\delta_k^m\}_{k=1}^{t-1} \right),$$

where $v^{ON} \left(x, \{\Delta_k^{\mathfrak{A}_t}\}_{k=t}^T; \{d_k^m\}_{k=1}^t, \{\delta_k^m\}_{k=1}^{t-1} \right)$ is the peak demand under the algorithm $\mathfrak{A}_t$ when the realization of $\mathbf{d}$ equals $x \in \mathcal{D}_t$.

Definition 2 At time slot t , the shrinking-horizon CR of a shrinking-horizon online PDM algorithm $\mathfrak{A}_t$ is

$$\pi_t^{\mathfrak{A}_t} = \max_{x \in \mathcal{D}_t} \frac{v^{ON} \left(x, \{\Delta_k^{\mathfrak{A}_t}\}_{k=t}^T; \{d_k^m\}_{k=1}^t, \{\delta_k^m\}_{k=1}^{t-1} \right)}{v^{OPT}(\mathbf{x})}.$$

Accordingly, the optimal shrinking-horizon CR π_t^* refers to the smallest CR that can be attained by a shrinking-horizon online PDM algorithm at time slot t .

The optimal shrinking-horizon CR π_t^* cannot be pre-computed like the best CR π^* , because it depends on the real-time information $(c_t, \{d_k^m\}_{k=1}^t, \{\delta_k^m\}_{k=1}^{t-1})$, apart from the prior information $(T, \underline{d}, \bar{d}, \bar{\delta})$. The optimal shrinking-horizon CR π_t^* quantifies the smallest possible gap between the online decision-making for the remaining time horizon $[T] \setminus [t-1]$ and the clairvoyant offline optimum for the whole horizon. The gap is due to the remaining uncertainty $\mathcal{D}_t$, realized net loads $\{d_k^m\}_{k=1}^t$, and current peak demand $\max_{k \in [t-1]} (d_k^m - \delta_k^m)$.

Proposition 4 At time slot t , the functions $\{\Delta_k^{\pi^*}\}_{k=t}^T$ of the pCR-PDM(π) algorithm is a shrinking-horizon online PDM algorithm if and only if $\pi \geq \pi_t^*$.

Algorithm 2: Real-time optimal pCR-PDM

Parameters: $c = c_1, T, \bar{\delta}, [\underline{d}, \bar{d}]$;
Input: $\pi > 1$ and a realization of $\mathbf{d}$;
Output: A realization of δ ;

- 1 **for** $t = 1, 2, \dots, T$ **do**
- 2 Construct $\mathbf{d}_t$ from measured inputs $\{d_k^m\}_{k \in [t]}$;
- 3 Compute π_t^* ;
- 4 $\delta_t = \lceil d_t^m - \pi_t^* v^{OPT}(\mathbf{d}_t) \rceil^+$;
- 5 Update $c_{t+1} = c_t - \delta_t^m$;

Algorithm 3: A bisection method for π_t^* in the real-time optimal pCR-PDM algorithm

Parameters: $T, \bar{\delta}, [\underline{d}, \bar{d}]$;
Input: $t, c_t, \{d_k^m\}_{k=1}^t, \{\delta_k^m\}_{k=1}^{t-1}$, and π_{t-1}^* ;
Output: π_t^* ;

- 1 $\pi_{lb} = \max_{k \in [t-1]} (d_k^m - \delta_k^m) / v^{OPT}(\mathbf{d}_t)$,
 $\pi_{ub} = \pi_{t-1}^*, q = \max_{\mathcal{I} \in \mathcal{I}_t} \text{AOCR-THR}(\pi_{lb}, \mathcal{I})$;
- 2 **if** $q \leq c_t$ **then**
- 3 $\pi_t^* = \pi_{lb}$, **return**;
- 4 **else**
- 5 **while** $\pi_{ub} - \pi_{lb} \geq \epsilon$ **do**
- 6 $\pi = (\pi_{lb} + \pi_{ub}) / 2$,
 $q = \max_{\mathcal{I} \in \mathcal{I}_t} \text{AOCR-THR}(\pi, \mathcal{I})$;
- 7 **if** $q > c_t$ **then**
- 8 $\pi_{lb} = \pi$;
- 9 **else**
- 10 $\pi_{ub} = \pi$;

Proposition 4 follows from Proposition 2, Lemma 1, and Theorem 1. It shows that the optimal shrinking-horizon CR characterizes the best competitiveness that can be achieved by making tactful online actions for the remaining horizon. Inspired by the rolling-optimization framework in MPC and optimal shrinking-horizon CRs, we extend the pCR-PDM(π^*) algorithm by rolling-optimizing the shrinking-horizon CR and determining δ_t by

$$\Delta_t^{\pi_t^*} \left(d_t^m; \{d_k^m\}_{k=1}^{t-1}, \{\delta_k^m\}_{k=1}^{t-1} \right) = \left[d_t^m - \pi_t^* v^{OPT}(\mathbf{d}_t) \right]^+,$$

for each $t \in [T]$. This extension is illustrated in Figure 2 and leads to the real-time optimal pCR-PDM algorithm with its pseudocodes presented in Algorithm 2. The real-time optimal pCR-PDM algorithm is a dynamic feedback policy affine over the synthetic state $v^{OPT}(\mathbf{d}_t)$, generalizing the static one – pCR-PDM(π^*).

The theorem below, proven in Appendix F, states that the real-time optimal algorithm retains the global optimal worst-case performance regarding CR.

Theorem 3 The real-time optimal pCR-PDM algorithm is feasible, and its CR is the best CR π^* for online PDM.

Apart from the optimal robustness regarding CR, we wonder how the real-time optimal pCR-PDM algorithm improves efficiency versus the pCR-PDM(π^*) algorithm. With $\pi^* = \pi_0^*$, the following corollary specifies the non-increasing property of the rolling-optimized shrinking-horizon CRs in Algorithm 2.

Corollary 1 *In the real-time optimal algorithm,*

$$\pi^* = \pi_0^* \geq \pi_1^* \geq \dots \geq \pi_T^*.$$

Corollary 1 follows from Proposition 4. By Theorem 3, we have $\pi_t^* = \pi^*$ for all $t \in \llbracket T \rrbracket$ in the worst case. However, in general cases, the inequality $\pi_\tau^* < \pi_{\tau-1}^*$ holds for certain $\tau \in \llbracket T \rrbracket$. That is, the real-time optimal pCR-PDM algorithm attains smaller online-to-offline ratios for non-worst-case realizations of $\mathbf{d}$. Thus, the dynamic policy (real-time optimal pCR-PDM) improves the general-case performance compared with its static counterpart (pCR-PDM(π^*)). The question arises of how to compute the optimal shrinking-horizon CRs at Line 3 of Algorithm 2.

4.2 Computing the optimal shrinking-horizon CR

Considering the realized inputs and committed actions makes it harder to compute the optimal shrinking-horizon CRs than the best CR. Nevertheless, Corollary 1 and the following observations help.

- Note that $\pi_t^* \geq \max_{k \in \llbracket t-1 \rrbracket} (d_k^m - \delta_k^m) / v^{OPT}(\mathbf{d}_t)$, because the online peak demand with $\mathbf{d}_t$ as the realization of $\mathbf{d}$ is no less than $\max_{k \in \llbracket t-1 \rrbracket} (d_k^m - \delta_k^m)$.
- It is unnecessary to decrease the current demand to be less than the peak demand over the past time slots.

Thus, a bisection method can be used to search for π_t^* . Define a linear program AOCR-THR($\pi, \mathcal{I}$) parameterized by $\pi \in [\max_{k \in \llbracket t-1 \rrbracket} (d_k^m - \delta_k^m) / v^{OPT}(\mathbf{d}_t), \pi_{t-1}^*]$ and $\mathcal{I} \in \mathcal{I}_t$, where $\mathcal{I}_t = \{\llbracket k \rrbracket \setminus \llbracket t \rrbracket \mid k = t+1, \dots, T\}$ as

$$\begin{aligned} & \max_{\mathbf{x} \in \mathcal{D}_t, u_i, \delta_{ij}} \sum_{i \in \mathcal{I}} (x_i - \pi u_i) + \\ & \left[d_t - \max \left\{ \pi v^{OPT}(\mathbf{d}_t), \max_{k \in \llbracket t-1 \rrbracket} (d_k^m - \delta_k^m) \right\} \right]^+ \\ & \text{subject to} \quad (6) - (9); \\ & \quad \max_{k \in \llbracket t-1 \rrbracket} (d_k^m - \delta_k^m) \leq \pi u_i, \forall i \in \mathcal{I}. \end{aligned}$$

The above objective refers to the aggregate amount of energy storage allocated over the set $\mathcal{I} \cup \{t\}$ of time slots assuming to uniformly apply Δ_t^π of pCR-PDM(π) to map d_k^m to δ_k^m for all $k \in \llbracket T \rrbracket \setminus \llbracket t-1 \rrbracket$. Similar to CR-Compute($\mathcal{I}$), the constraints of AOCR-THR($\pi, \mathcal{I}$) are due to the offline PDM problem solved in each time slot in the real-time optimal pCR-PDM algorithm. The difference lies in that the demand for a future time slot is

never decreased to be less than the current peak demand, leading to the inequality constraints on the last line. By similar arguments for computing the best CR π^* , the optimal shrinking-horizon CR at time slot t should not make $\max_{\mathcal{I} \in \mathcal{I}_t} \text{AOCR-THR}(\pi, \mathcal{I})$ exceed the remaining capacity c_t . Thus, the bisection method for computing π_t^* in Algorithm 2 is obtained and presented in Algorithm 3.

Remark 4 *The state variable c_t plays a role in the real-time optimal pCR-PDM algorithm. This fact makes Algorithm 2 more adaptive to contingency, less sensitive to model mismatch, and thus more useful for practical applications than pCR-PDM(π^*). In practice, the remaining capacity c_t can be measured or estimated for each time slot t , rather than updated as shown in Algorithm 2.*

4.3 pCR-PDM(π^*) versus real-time optimal PDM

In summary, both the pCR-PDM(π^*) and real-time optimal pCR-PDM algorithms are guaranteed to attain the best CR π^* and require solving the same offline PDM problem for an assumed net load profile at each time slot. However, they differ in the following aspects.

- The pCR-PDM(π^*) algorithm consistently applies the best CR π^* in its control policy (static feedback policy). In contrast, the real-time optimal pCR-PDM dynamically adjusts its control policy using a sequence of optimal shrinking-horizon CRs that evolve over time (dynamic feedback policy).
- The real-time optimal pCR-PDM algorithm incorporates the state variable c_t and the peak usage observed up to the current time slot to compute the optimal shrinking-horizon CRs. This enables it to adapt more effectively to real-time changes in system conditions, unlike the static policy used in pCR-PDM(π^*).
- The pCR-PDM(π^*) algorithm is simpler to implement, but the real-time optimal pCR-PDM algorithm is expected to deliver superior general performance, as demonstrated in the simulation results later.

Similar to MPC, the real-time optimal pCR-PDM algorithm employs the rolling-horizon optimization framework. The work [19] applied an *algorithm-robustification* procedure to an MPC algorithm for the best CR regarding the original uncertainty $\mathcal{D}$ and good general-case performance. With optimal shrinking-horizon CRs, the algorithm-robustification procedure can be refined at each time slot for real-time optimal robustness.

5 Simulation Cases

5.1 Data and evaluation setup

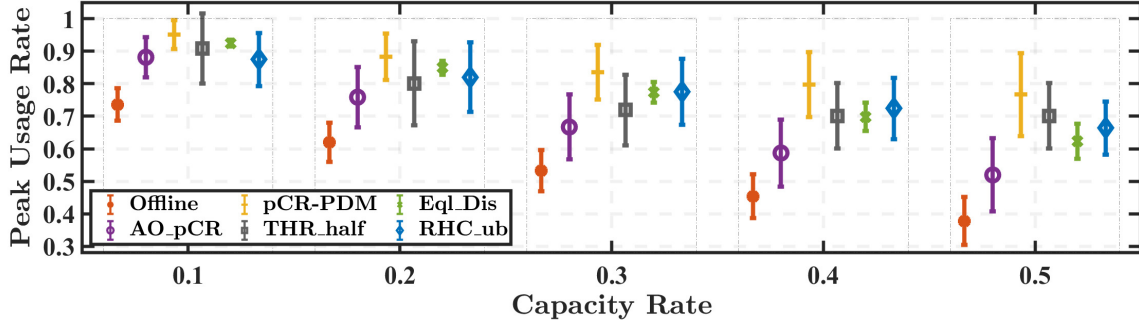


Figure 3. Performances of the optimal offline solution (*ref.* Offline), real-time optimal pCR-PDM (*ref.* AO_pCR), pCR-PDM(π^*) (*ref.* pCR-PDM), two threshold-based algorithms (*ref.* THR_half and Eql_Dis), and a shrinking RHC algorithm (*ref.* RHC_ub).

Table 2

The Online-to-Offline Ratios of Peak Demand

Capacity Rate	0.10	0.20	0.30	0.40	0.50
π^*	1.37	1.46	1.60	1.80	2.08
pCR-PDM	1.29	1.42	1.57	1.75	2.02
AO_pCR	1.20	1.22	1.25	1.29	1.37

Simulations are based on 3-month traces from an EV charging station. A collection of transactions constitute the raw data. Each transaction records the starting time, duration (in minutes), and consumed energy (in kWh) of an ordered charging. Assume that the charging rate is constant in the recorded duration and convert the raw data into a time series regarding the electricity consumption of the station. Then, the on-peak horizon in a day is identified and uniformly divided into 15-minute time slots. Specifically, set $T = 20$ and $(\underline{d}, \bar{d}) = (442.91, 1020.10)$ kWh.

5.2 Real-time optimal pCR-PDM versus pCR-PDM(π^*)

Define the capacity rate as the ratio of the capacity and average daily energy consumption in the on-peak horizon, and define the empirical performance ratio of an algorithm as the ratio of the average peak demand under the algorithm and that under the optimal offline solution. In the typical setup from real-world traces, Table 2 records the best CRs (namely π^*) and the empirical performance ratios of the pCR-PDM(π^*) (*ref.* pCR-PDM) and real-time optimal pCR-PDM algorithms (*ref.* AO_pCR) under different capacities. As seen, the real-time optimal pCR-PDM consistently attains smaller empirical performance ratios compared with pCR-PDM(π^*). Also, its performance ratios are much smaller than the corresponding best CRs.

Figure 3 also shows that the real-time optimal pCR-PDM algorithm attains better empirical performance than pCR-PDM(π^*). Define the peak usage rate as the ratio between the peak demand and the peak net load. The average peak usage rates and the corresponding standard deviations are delineated in Figure 3. It shows that,

on average, the peak demand reduction under the real-time optimal pCR-PDM algorithm is more than twice that under pCR-PDM(π^*), while the peak usage rate under the real-time optimal pCR-PDM algorithm decreases faster than that under pCR-PDM(π^*) as the capacity increases. Notably, when the capacity rate is 30%, the real-time optimal algorithm with unknown future inputs achieves up to 77% of the peak demand reduction under the optimal offline outcomes where complete knowledge is known beforehand.

Overall, these observations substantiate the significance of exploiting real-time information and the advantage of the real-time optimal pCR-PDM over pCR-PDM(π^*).

5.3 Comparison with baseline alternatives

Two classes of conceivable alternatives are considered: threshold-based and shrinking RHC algorithms.

- The first class consists of two threshold-based algorithms. In one algorithm (*ref.* THR_half), the demand threshold is set as the mean $(\underline{d} + \bar{d})/2$ of demand bounds, and the storage is consumed such that the demand reduces to the threshold or the storage runs out. The other algorithm (*ref.* Eql_Dis), maintains δ_t as c/T .
- The second class is a shrinking RHC algorithm that has an accurate look-ahead window of size as a quarter of the on-peak horizon and assumes each future demand beyond the window as the upper bound $\bar{d}$ (*ref.* RHC_ub)¹.

Figure 3 shows that the real-time optimal pCR-PDM attains much better empirical performance than the threshold and RHC algorithms under a wide range of capacities. When the capacity rate is 30%, the real-time optimal pCR-PDM can improve the average peak

¹ If the window size reduces to one, the RHC algorithm can be regarded as a robust algorithm concerning the absolute worst-case performance.

demand reduction by at least 19% compared to the alternatives. As the capacity increases, the average peak usage rate of the real-time optimal pCR-PDM decreases much faster than that of each baseline algorithm.

Remark 5 *The pCR-PDM(π^*) algorithm often performs worse than the three alternative algorithms. That is not surprising because many robust algorithms like pCR-PDM(π^*) are criticized for only focusing on the worst-case performance over the original uncertainty set and hardly concerning the general-case performance.*

Remark 6 *Two other threshold-based algorithms are evaluated. One sets the demand threshold as the average optimal offline peak demand over the recorded days, and the other maintains the ratio δ_t/d_t as the capacity rate. Also, RHC algorithms with different assumptions of future demands are evaluated, including using $\underline{d}$ and the mean $(\underline{d} + \bar{d})/2$ of net demand bounds. Their performances are similar to or worse than the reported baselines and thus excluded for brevity.*

6 Conclusions

This work delves into the online peak-demand minimization with in-house energy storage consumption. The resulting mathematical problem is theoretically challenging yet practically relevant across broad applications. Within the primal-based CR-Pursuit framework, we developed a series of online algorithms with specified CRs and used linear programs to compute the best CR.

Inspired by the rolling-optimization framework in MPC, we extended the CR concept to shrinking-horizon CRs. This extension enabled the development of a primal-based real-time decision-making scheme that adaptively incorporates the real-time information, thereby improving the original CR-Pursuit framework. This scheme achieves real-time optimal robustness by rolling-optimizing shrinking-horizon CRs, offering a promising pathway for integrating robustness and efficiency. Apart from theoretical justifications, we empirically validated the efficacy of the proposed approach through simulations based on real-world traces. Overall, this work advances the understanding of real-time peak shaving and energy management within the context of competitive analysis in the control community.

Despite its strengths, the proposed approach faces several challenges concerning its generalizability and practical implementations. First, the application of the proposed approach is problem-specific, which involves the computation of shrinking-horizon CRs and the method for maintaining the computed CRs. Particularly, obtaining shrinking-horizon CRs is often computationally intensive. Second, although our method can be readily modified to absorb short-term predictions [1], achieving optimal performance in this context remains a significant

challenge, particularly when such look-ahead information is from potentially unreliable machine-learned predictions [40]. Third, further research is needed to systematically compare and potentially unify the algorithm-robustification techniques in [3] with our real-time optimal framework. Moreover, it is unclear how to make the most of the proposed approach for the scenarios involving both demand and volume charges.

Furthermore, our analysis is based on a simplified setting, and several practical considerations remain unaddressed. These include determining how to obtain model parameters from a real system, integrating the energy storage system that may be replenishable in the on-peak horizon, and accounting for the degradation costs. Addressing these challenges will require additional techniques and refinements. Nevertheless, the proposed real-time decision-making framework demonstrates strong potential for broader applicability and serves as a promising foundation for future research.

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A Proof of Proposition 1

The Weierstrass extreme value theorem establishes the existence of the optimal solution to the offline PDM. If $v^* = 0$, the proof is trivial. Next, assume $v^* > 0$. By the capacity constraint, the optimal objective value of the offline PDM should be no less than u .

If $u \geq \max_{t \in [T]} d_t - \bar{\delta}$, then δ^* is a feasible solution achieving the peak demand u because

$$\begin{aligned} d_t - \delta_t^* &= d_t - [d_t - u]^+ \leq u, \\ \sum_{t=1}^T \delta_t^* &= c, \quad \delta_t^* \leq d_t - u \leq d_t, \quad \text{and} \\ \delta_t^* &\leq d_t - u \leq \max_{t \in [T]} d_t - u \leq \bar{\delta}. \end{aligned}$$

Thus, the optimal offline peak demand is u and δ^* is an optimal solution. If $u < \max_{t \in [T]} d_t - \bar{\delta}$, the optimal offline peak demand is $\max_{t \in [T]} d_t - \bar{\delta}$ and δ^* is a feasible solution achieving this peak demand because

$$\begin{aligned} \max_{t \in [T]} (d_t - \delta_t^*) &= \max_{t \in [T]} d_t - \bar{\delta}, \\ \sum_{t=1}^T \delta_t^* &< c, \quad \max_{t \in [T]} \delta_t^* = \bar{\delta} \quad \text{and} \\ \delta_t^* &\leq d_t - \left(\max_{t \in [T]} d_t - \bar{\delta} \right) \leq d_t. \end{aligned}$$

B Proof of Proposition 2

The necessity follows from (1), and it remains to prove the sufficiency. It follows from $\Phi(\pi) \leq c$ and $\Delta_t^\pi(d_t^m; \{d_k^m\}_{k=1}^{t-1}, \{\delta_k^m\}_{k=1}^{t-1}) \geq 0$ for all $t \in \llbracket T \rrbracket$ that

$$\Delta_t^\pi(d_t^m; \{d_k^m\}_{k=1}^{t-1}, \{\delta_k^m\}_{k=1}^{t-1}) \leq c_t, \text{ for all } t \in \llbracket T \rrbracket.$$

It follows from $d_t^m - v^{OPT}(\mathbf{d}_t) \leq \bar{\delta}$ that

$$\Delta_t^\pi(d_t^m; \{d_k^m\}_{k=1}^{t-1}, \{\delta_k^m\}_{k=1}^{t-1}) \leq \bar{\delta}, \text{ for all } t \in \llbracket T \rrbracket.$$

It follows from $d_t^m \geq 0$ and $v^{OPT}(\mathbf{d}_t) \geq 0$ that

$$\Delta_t^\pi(d_t^m; \{d_k^m\}_{k=1}^{t-1}, \{\delta_k^m\}_{k=1}^{t-1}) \leq d_t^m, \text{ for all } t \in \llbracket T \rrbracket.$$

Thus, the online constraint (4) is satisfied for all possible realizations, which completes the proof.

C Proof of Lemma 1

The function $f(\pi) = [d_t^m - \pi v^{OPT}(\mathbf{d}_t)]^+$ is strictly decreasing over π when its function value is positive. Then, summarizing the aforementioned functions over $t \in \llbracket T \rrbracket$ gives another nonincreasing function over π : $\sum_{t=1}^T f_t(\pi)$, which is also strictly decreasing when its function value is positive. It follows that $\Phi(\pi)$ is strictly decreasing before it attains zero.

D Proof of Theorem 1

Assumption 1 implies that $\Phi(0) > c$. Then, the uniqueness of the solution to $\Phi(\pi) = c$ follows from Lemma 1. Let $\delta^\mathfrak{A}$ and δ^{π^*} respectively be the output sequences of an online PDM algorithm $\mathfrak{A}$ and the pCR-PDM(π^*) algorithm. Let $\hat{\mathbf{d}}$ be a worst-case realization of $\mathbf{d}$ under the pCR-PDM(π^*) algorithm.

From $t = 1$, set $d_t^m = \hat{d}_t$ until $\delta_t^\mathfrak{A} < \delta_t^{\pi^*}$ or $\delta_t^\mathfrak{A} > 0$ and realize the remaining net loads by $\hat{\mathbf{d}}$. One of the two situations will happen since pCR-PDM(π^*) uses up the energy storage under the realization $\hat{\mathbf{d}}$. If the former happens, it is clear that the CR of $\mathfrak{A}$ is greater than π^* . Otherwise, the CR of $\mathfrak{A}$ is no less than π^* . Thus, no online PDM algorithm has a CR which is strictly smaller than π^* . Further, by Proposition 2, the proof is completed.

E Proof of Lemma 3

Let $\hat{\mathbf{d}}$ be a worst-case realization of $\mathbf{d}$ under the pCR-PDM(π^*) algorithm. If the associated output sequence

is described as in Lemma 3, then the proof is completed. Otherwise, there exists an index i such that

$$\begin{aligned} \hat{d}_i - \pi v^{OPT}(\mathbf{d}_i) &\leq 0 \text{ and} \\ \hat{d}_{i+1} - \pi v^{OPT}(\mathbf{d}_{i+1}) &> 0. \end{aligned}$$

It follows that $\hat{d}_i < \hat{d}_{i+1}$. Construct a new realization $\tilde{\mathbf{d}}$ from $\hat{\mathbf{d}}$ by exchanging $\hat{d}_i$ and $\hat{d}_{i+1}$. Applying pCR-PDM(π^*) algorithm for the realization $\tilde{\mathbf{d}}$, we obtain

$$\begin{aligned} \tilde{d}_i - \pi v^{OPT}(\mathbf{d}_i) &> 0 \text{ and} \\ \tilde{d}_{i+1} - \pi v^{OPT}(\mathbf{d}_{i+1}) &\leq 0. \end{aligned}$$

It follows that $\delta_i^m > 0$, $\delta_{i+1}^m > 0$, and $\delta_i^m + \delta_{i+1}^m$ is no less than that with the previous realization $\hat{\mathbf{d}}$. Note that δ_i^m for all $k \in \llbracket T \rrbracket \setminus \{i, i+1\}$ remains the same. Since $\hat{\mathbf{d}}$ is a worst-case realization, the new realization $\tilde{\mathbf{d}}$ is also in the worst case. Following similar arguments, by a series of exchanges of adjacent elements, we can obtain a worst-case realization as described in the lemma.

F Proof of Theorem 3

The feasibility of the real-time optimal pCR-PDM algorithm is proven by repeatedly applying Proposition 4 and the definitions of optimal shrinking-horizon CRs. If the input sequence $\mathbf{d}$ is in the worst-case regarding pCR-PDM(π^*), then $\pi_t^* = \pi^*$, for all $t \in \llbracket T \rrbracket$; otherwise $\pi_t^* \leq \pi^* \leq \pi^*$. Thus, it follows from the definition that the CR of the real-time optimal pCR-PDM algorithm is π^* .

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